

Automation at Scale Across Domains

Data Center/Campus/WAN

AUTOCON S

THE NETWORK AUTOMATION CONFERENCE



\$ whoami

- Daytime: Global IT Network Architect - Rohde & Schwarz GmbH & Co. KG
- Nighttime: VyOS Universal Router - core maintainer

What gpt-oss-120b describes me:

I combine deep routing and switching expertise with a decade of embedded Linux experience and eight years as a VyOS core maintainer, delivering secure, automated, and scalable network solutions that move from code to production in minutes instead of weeks.

ONE COMPANY, THREE DIVISIONS, DIVERSE MARKETS: WE ARE A RELIABLE TECHNOLOGY PARTNER

TEST & MEASUREMENT



Wireless | Industry, Components &
Research | Aerospace & Defense Testing |
Automotive

TECHNOLOGY SYSTEMS



Mobile Network Testing | Secure
Communications | Critical Infrastructure |
Government | IP Network Analytics |
Broadcast, Amplifiers & Media

NETWORKS & CYBERSECURITY



Cloud Managed Business |
Network & Security Solutions |
Certified & High-Grade Crypto Solutions



Building an Automation Playground

Where our journey begins

- Pain points started to increase rapidly in Q1/2020
- Single vendor, but
 - Multiple Operating Systems
 - Multiple CLI variants
- Missing automation framework
- Increasing demand on port density and speed
- No dedicated QA lab environment
- Who is this new EVPN-VXLAN technology?

Scouting a future-proof NOS

- Vendor-provided and vendor-supported automation framework
 - Ideally open-source, so we can tailor it to our needs
 - Extend in the open-source spirit
- Vendor-supported virtualization images for QA
 - Fail-fast and iterate in a simulated environment
 - Break things on purpose!
- Simulation support for EVE-NG
 - There was no Containerlab back in 2020

Scouting a future-proof NOS #2

- Arista provided image download access for a POC
 - Set up a **CloudVision** POC cluster on our hypervisors
 - Used vEOS-lab so we could get to know each other
- Moved vEOS-lab VMs into **EVE-NG**
- Recreated current WAN topology using **Arista AVD**
- Reiterate the current design and future optimisation
 - Seamless changes due to the virtual QA environment
- One source for both PROD and QUAL environment (git)



Putting pieces together

- Design, deploy and connect two EVPN-VXLAN fabrics
- Set up everything in EVE-NG
 - maintain a 100% replica of the network at all times
- Test network convergence behavior/routing changes
- Have the full configuration done
 - ... while waiting on the hardware

“There won’t be much
Day-2 work” - or so we
thought

Expanding from WAN and DC to CAMPUS

- “Once the network is up and running, we will only have a few changes, e.g., adding one or two VLANs.”
- New features bring new flexibility – and customers are requesting new L2/L3 domains at a high rate
- A “slight” increase in the expected change rate
 - 2021 (109), 2022 (280), 2023 (647), 2024 (983), 2025 (1060)

Change Control

Manage, review, and execute change controls

● Live Time ⓘ

+ Create

🔍 Filter by change control name, ID o...

All 1295 🔗 Pending Approval 0 👤 Approved 0 ⚙ Running 0 ❌ Failed 235 ✅ Success 1060

Date Range ⓘ Jan 1, 2025 → Dec 31, 2025 🗑 ↺

🔼 Device Filter (show all)

Why automate across domains?



Business Drivers

- Reliability - We act as an internal service provider, and this is the company's core backbone.
- Speed - New customer demands arrive frequently
- Consistency - We have all forgotten to add a VLAN to a Port-Channel at least once
- MTTR - Mean Time to Repair: thanks to Git, we can always revert to a known-good state

Human Drivers

- We tend to be lazy, and laziness is a good thing – it drives automation
- Nobody likes siloed config templates
- Configuration drift: “Just one little change... and one more tiny thing.”
- Unpredictable change windows
- Lack of a QA environment

Declarative Automation Model

Architecture Highlights

- Git is used for both source code and documentation (MD)
- NetBox as an additional source of truth alongside Git
- GitLab is also used for CI/CD runs and pushes to Arista CloudVision
- Use ansible-lint and yamllint to keep everything consistent
- ANTA to detect issues
 - E.g., decommissioned endpoint ports



Data-driven design

- Data determines what to render or include in the resulting switch configuration
 - VLANs on Port-Channels
 - VLAN database
 - VNI (VXLAN Network Identifiers)
- Source of truth (NetBox)
- Idempotence is a must

Core Concepts

- Define once, reuse often (Encapsulation)

```
(08:32) breuni_c MU22222:~/devnetops [master] # cat group_vars/all.yml | sed '/^#/d' | wc -l  
797
```

- Structured Inventory: YAML and NetBox (Abstraction)
 - One inventory per domain: DC, LAN and WAN
 - One playbook per site and domain: MU DC, MU LAN or MU WAN
- Domain-specific extension: DC.yml, LAN.yml and WAN.yml (Inheritance)
- Who changed what, why and when (Audit Trail via git blame)

Domain Specific Add-Ons

- EVPN active-active multihoming ESI IDs
- L3 edge point-to-point configurations
- DCI underlay
- Maintenance Mode profiles
- ACLs
- DNS/NTP settings with generic all.yml fallback

GitLab CI Pipeline

✓ Dockerfile
Build OCI



✓ Deploy OCI
Post Build



✓ MD concepts to PDF
Post Build



✓ YAML Lint
Post Build



✓ [redacted]:qual
Generate switch config



✓ [redacted]:qual
Generate switch config



✓ [redacted]:qual
Generate switch config



✓ [redacted]:qual
Generate switch config



✓ [redacted]:qual
Generate switch config



✓ MU DC:qual
Generate switch config



✓ MU LAN:prod
Generate switch config



✓ [redacted]:qual
Generate switch config



✓ [redacted]:qual
Generate switch config



✓ [redacted]:qual
Generate switch config



✓ WAN:qual
Generate switch config



✓ [redacted]:prod
Generate switch config



✓ [redacted]:prod
Generate switch config



✓ [redacted]:prod
Generate switch config



✓ [redacted]:prod
Generate switch config



✓ MU DC:prod
Generate switch config



✓ [redacted]:prod
Generate switch config



✓ [redacted]:prod
Generate switch config

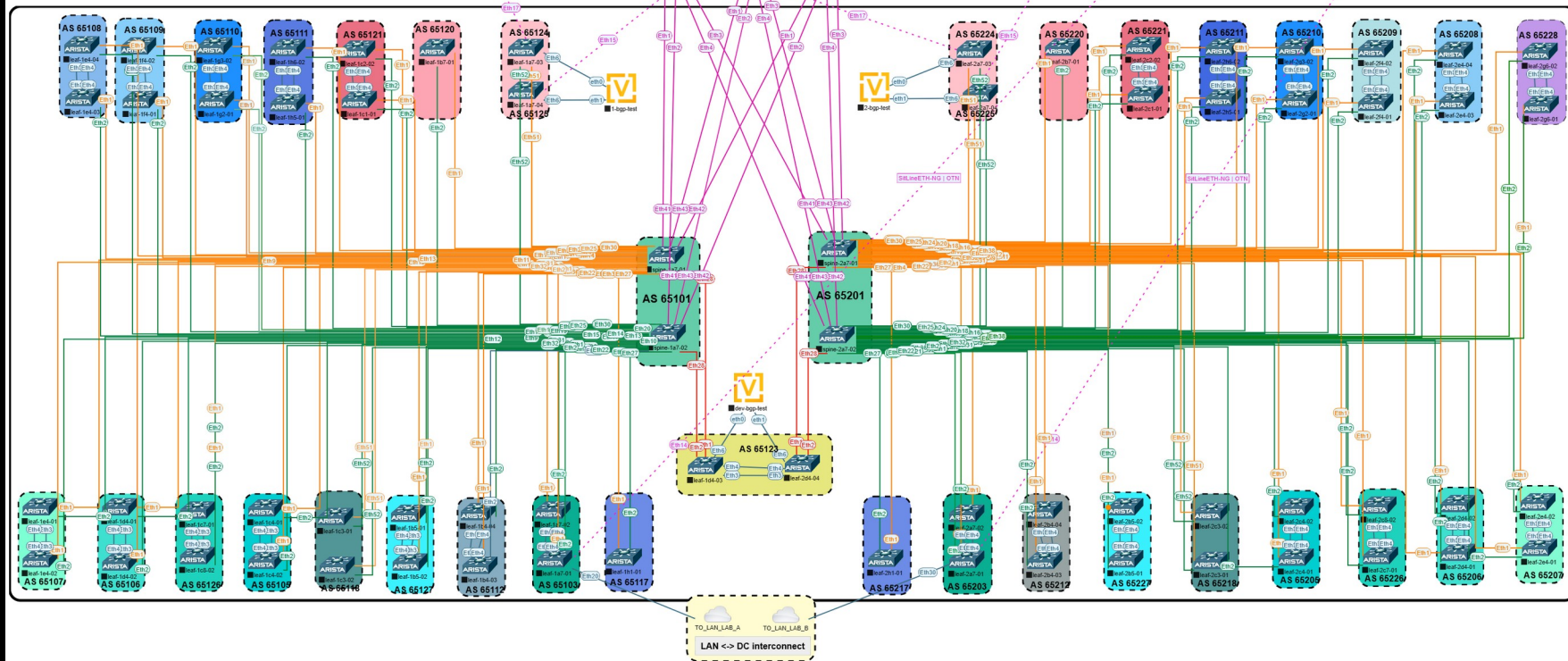
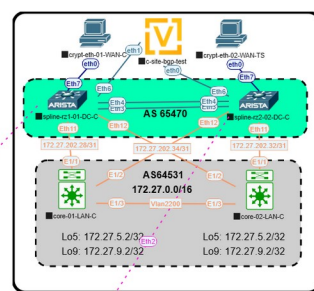
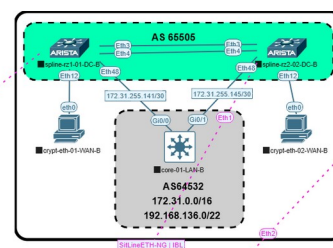
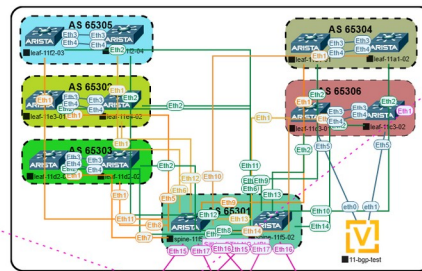
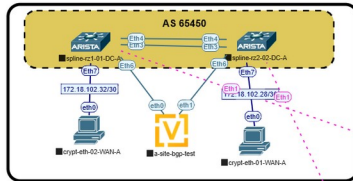


✓ [redacted]:prod
Generate switch config



✓ WAN:prod
Generate switch config





Scaling Tips

- 2-byte ASN vs 4-byte ASN (BGP RD field is only 32-bit)
- IP addresses for MLAG peer links
 - choose wisely, e.g., 169.254.0.0/31. Renumbering is painful
 - They could conflict with customer IPs in a customer VRF
- Automated documentation
- GitLab runners
- DC, Campus, WAN – you name it. They are all just tenants on our network
- Think BIG

Lessons Learned

- “Not many Day-2 operations” turned into a fast-evolving, service-provider-like network
- Always test changes in a QA environment – even the smallest change can open the gates to STP hell
- We spend 80% of our time in the QA environment – use proper servers
- Do not forget legacy 100 Mbps equipment SFP-10G-MRA-T
- Ansible AVD evolves quickly – update frequently to save time in the long run



Questions?